

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.

4. Determine the remaining unused IP address range within the given network block.

Department	Subnet Mask	Network Address	First Host	Last Host	Broadcast Address	Usable Hosts
School of Computing	/25 (255.255.255.128)	192.168.100.0	192.168.100.1	192.168.100.126	192.168.100.127	126
School of Electronics	/26 (255.255.255.192)	192.168.100.128	192.168.100.129	192.168.100.190	192.168.100.191	62
School of Mechanical Engineering	/27 (255.255.255.224)	192.168.100.192	192.168.100.193	192.168.100.222	192.168.100.223	30

Calculations

1. School of Computing (/25)

- Subnet Mask: 255.255.255.128
- Block Size: 128 addresses
- Network Address: 192.168.100.0
- Broadcast Address: 192.168.100.127
- Valid Host Range: 192.168.100.1 – 192.168.100.126
- Usable Hosts: $2^{32-25} - 2 = 128 - 2 = 126$

2. School of Electronics (/26)

- **Subnet Mask:** 255.255.255.192
- **Block Size:** 64 addresses
- **Network Address:** 192.168.100.128
- **Broadcast Address:** 192.168.100.191
- **Valid Host Range:** 192.168.100.129 – 192.168.100.190
- **Usable Hosts:** $2^{32-26} - 2 = 64 - 2 = 62$

3. School of Mechanical Engineering (/27)

- **Subnet Mask:** 255.255.255.224
- **Block Size:** 32 addresses
- **Network Address:** 192.168.100.192
- **Broadcast Address:** 192.168.100.223
- **Valid Host Range:** 192.168.100.193 – 192.168.100.222
- **Usable Hosts:** $2^{32-27} - 2 = 32 - 2 = 30$

Remaining Unused IP Address Range

After allocating the three subnets:

- **Computing:** 192.168.100.0 – 192.168.100.127
- **Electronics:** 192.168.100.128 – 192.168.100.191
- **Mechanical:** 192.168.100.192 – 192.168.100.223

The remaining unused portion of the original 192.168.100.0/24 network is:

- **Unused Network Range:** 192.168.100.224 – 192.168.100.255
- This corresponds to a /27 subnet.
- **Network Address:** 192.168.100.224
- **Broadcast Address:** 192.168.100.255
- **Valid Host Range:** 192.168.100.225 – 192.168.100.254
- **Usable Hosts Available:** 30

Final Answer (Summary)

Department	Network Address	Broadcast Address	Host Range	Usable Hosts
School of Computing	192.168.100.0/25	192.168.100.127	192.168.100.1 – 192.168.100.126	126
School of Electronics	192.168.100.128/26	192.168.100.191	192.168.100.129 – 192.168.100.190	62
School of Mechanical Engineering	192.168.100.192/27	192.168.100.223	192.168.100.193 – 192.168.100.222	30
Unused Range	192.168.100.224/27	192.168.100.255	192.168.100.225 – 192.168.100.254	30

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require

different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

Solution

Department	Subnet Mask	Network Address	First Host	Last Host	Broadcast Address	Usable Hosts
Software Development	/24 (255.255.255.0)	10.10.0.0	10.10.0.1	10.10.0.254	10.10.0.255	254
Quality Assurance	/25 (255.255.255.128)	10.10.1.0	10.10.1.1	10.10.1.126	10.10.1.127	126
Human Resources	/26 (255.255.255.192)	10.10.1.128	10.10.1.129	10.10.1.190	10.10.1.191	62
Executive Management	/27 (255.255.255.224)	10.10.1.192	10.10.1.193	10.10.1.222	10.10.1.223	30

Calculations

1. Software Development (/24)

- Subnet Mask: 255.255.255.0
- Network Address: 10.10.0.0
- Broadcast Address: 10.10.0.255
- Valid Host Range: 10.10.0.1 – 10.10.0.254
- Usable Hosts: $2^{32-24} - 2 = 256 - 2 = 254$

2. Quality Assurance (/25)

- Subnet Mask: 255.255.255.128
- Network Address: 10.10.1.0
- Broadcast Address: 10.10.1.127
- Valid Host Range: 10.10.1.1 – 10.10.1.126
- Usable Hosts: $2^{32-25} - 2 = 128 - 2 = 126$

3. Human Resources (/26)

- Subnet Mask: 255.255.255.192
- Network Address: 10.10.1.128
- Broadcast Address: 10.10.1.191
- Valid Host Range: 10.10.1.129 – 10.10.1.190
- Usable Hosts: $2^{32-26} - 2 = 64 - 2 = 62$

4. Executive Management (/27)

- Subnet Mask: 255.255.255.224
- Network Address: 10.10.1.192
- Broadcast Address: 10.10.1.223
- Valid Host Range: 10.10.1.193 – 10.10.1.222
- Usable Hosts: $2^{32-27} - 2 = 32 - 2 = 30$

Remaining Unused IP Address Range

Allocated subnets use:

- 10.10.0.0 – 10.10.0.255 (/24)
- 10.10.1.0 – 10.10.1.127 (/25)
- 10.10.1.128 – 10.10.1.191 (/26)
- 10.10.1.192 – 10.10.1.223 (/27)

The remaining unused addresses are:

- 10.10.1.224 – 10.10.255.255

This includes:

- The remaining /27 block from 10.10.1.224 – 10.10.1.255 (32 addresses), and
- All addresses from 10.10.2.0 – 10.10.255.255 within the original 10.10.0.0/16 network.

Final Answer (Summary)

Department	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
Software Development	10.10.0.0/24	10.10.0.255	10.10.0.1 – 10.10.0.254	254
Quality Assurance	10.10.1.0/25	10.10.1.127	10.10.1.1 – 10.10.1.126	126
Human Resources	10.10.1.128/26	10.10.1.191	10.10.1.129 – 10.10.1.190	62
Executive Management	10.10.1.192/27	10.10.1.223	10.10.1.193 – 10.10.1.222	30
Unused Range	10.10.1.224 – 10.10.255.255	—	—	Remaining addresses within the 10.10.0.0/16 block

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.

2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

1. Convert the IPv6 Address into Compressed Notation

Rules:

- Remove leading zeros in each hexadecimal block.
- Replace one consecutive sequence of all-zero blocks with `::`.

Compressed IPv6 Address

2001:db8:abcd::/64

2. Expand the Compressed Address Back into Full Notation

Expanding `2001:db8:abcd::` gives:

2001:0db8:abcd:0000:0000:0000:0000/64

3. Identify the Network Prefix (/64)

A /64 prefix means:

- The **first 64 bits** represent the network portion.
- The **last 64 bits** represent the interface (host) identifier.

Network Prefix

2001:0db8:abcd:0000::/64

or equivalently,

2001:db8:abcd:0::/64

Network Portion (64 bits):

```
2001:0db8:abcd:0000
```

Host Portion (64 bits):

```
0000:0000:0000:0000
```

4. Calculate the Number of Host Addresses

Host bits = $128 - 64 = 64$

Total possible host addresses:

$$2^{64} \\ = 18,446,744,073,709,551,616$$

Therefore,

Total Host Addresses = 18,446,744,073,709,551,616

5. Total Number of Possible Host Addresses Available

Since IPv6 does **not** reserve network and broadcast addresses like IPv4, all interface identifiers are available.

Total Possible Addresses:

18,446,744,073,709,551,616 (2^{64})

Final Answer (Summary)

Task	Answer
Given IPv6 Address	2001:0db8:abcd:0000:0000:0000:0000/64
Compressed Notation	2001:db8:abcd::/64
Expanded Notation	2001:0db8:abcd:0000:0000:0000:0000/64
Network Prefix (/64)	2001:db8:abcd:0::/64
Network Portion	2001:0db8:abcd:0000
Host Portion	0000:0000:0000:0000
Host Bits	64
Total Possible Host Addresses	$2^{64} = 18,446,744,073,709,551,616$

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

1. Determine the Destination Subnet

For a /24 subnet mask (255.255.255.0), the first three octets represent the network portion.

Destination IP:

192.168.2.45

Network Address:

192.168.2.0/24

Destination Subnet: 192.168.2.0/24

2. Matching Routing Table Entry

Compare the destination IP with the routing table:

Routing Table Entry	Match?
192.168.1.0/24	No
192.168.2.0/24	Yes
192.168.3.0/24	No

Matching Routing Table Entry: 192.168.2.0/24

3. Next-Hop Network/Interface

Since the destination belongs to **192.168.2.0/24**, the router forwards the packet through the interface (or next hop) connected to that subnet.

Next-Hop Network/Interface:

- **Network:** 192.168.2.0/24
- **Outgoing Interface:** The router interface configured for the **192.168.2.0/24** network (for example, **GigabitEthernet0/1**, depending on the router configuration).

4. Default Route (If No Match Exists)

If the destination IP does not match any routing table entry, the router forwards the packet using the **default route**.

The default route is:

0.0.0.0/0

This route directs unmatched packets to the default gateway or ISP.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

1. Administration LAN (192.168.10.0/26)

- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63
- Valid Host Range: 192.168.10.1 – 192.168.10.62
- Default Gateway: 192.168.10.1

Suggested IP Addresses

Computer	IP Address
PC 1	192.168.10.2
PC 2	192.168.10.3
PC 3	192.168.10.4

2. Finance LAN (192.168.10.64/26)

- Network Address: 192.168.10.64
- Broadcast Address: 192.168.10.127
- Valid Host Range: 192.168.10.65 – 192.168.10.126
- Default Gateway: 192.168.10.65

Suggested IP Addresses

Computer	IP Address
PC 1	192.168.10.66
PC 2	192.168.10.67
PC 3	192.168.10.68

LAN	Network Address	Broadcast Address	Valid Host Range	Default Gateway
Administration	192.168.10.0/26	192.168.10.63	192.168.10.1 – 192.168.10.62	192.168.10.1
Finance	192.168.10.64/26	192.168.10.127	192.168.10.65 – 192.168.10.126	192.168.10.65

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured,	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

		and logically presented			
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